

LITERATURE SURVEY

[1].Authors: S. Sankaranarayanan & A. Periasamy (2021) Smart Infrastructure System Pilot tested – institutional use This research by Sankaranarayanan and Periasamy (2021) presents a comprehensive and technologically advanced approach to mitigating suicides by hanging from ceiling fans, a tragic phenomenon often reported in domestic and institutional environments. Their design integrates a PIC16F877A microcontroller, a programmable chip known for reliability and cost-efficiency, with a combination of Passive Infrared (PIR) and flex sensors to detect abnormal motion or bending of the fan rod. The PIR sensor continuously monitors human activity beneath the fan, while the flex sensor measures changes in the rod's angle or pressure distribution, allowing the system to detect the onset of a hanging attempt almost instantaneously. When a critical load or static presence is identified, the system triggers a buzzer to provide an audible alert and simultaneously activates a GSM module, which transmits a distress message to pre-registered mobile numbers. The standout innovation lies in the actuator driven mechanical intervention: upon detection, a small motor-driven mechanism lowers or extends the fan rod, effectively reducing the hanging height and preventing neck compression. This dual action—alert and mechanical disruption—makes the system proactive rather than reactive. Their prototype demonstrated high sensitivity and reliability in testing scenarios, with minimal false triggers when calibrated correctly. However, the study notes limitations, such as dependence on continuous power supply and potential wear of mechanical components over time. Future recommendations include integrating IoT-based monitoring, battery backup, and data analytics for behavioral pattern recognition. Overall, this work contributes a holistic electromechanical design that merges automation, safety engineering, and communication technology to prevent suicides effectively in real world applications like hostels, hospitals, and homes.

[2].Suicide Avoidance System Based on Ceiling Fan Authors: Senthilkumar D. et al. (2020) Senthilkumar and colleagues (2020) designed a suicide avoidance mechanism emphasizing weight based sensing and relay-controlled mechanical action. Unlike designs relying on motion detection, this system directly measures the load applied to the ceiling fan rod. A precision load cell or strain gauge sensor detects when the fan experiences a sudden or sustained load exceeding a defined threshold (usually >20 kg). Once the threshold is crossed, the microcontroller instantly activates a relay circuit that triggers the rod extension mechanism—a telescopic system that extends downward, reducing the tension necessary to sustain a hanging body. An essential innovation of this model is its wireless alerting system, which uses

GSM-based SMS notifications or IoT network integration to alert nearby caregivers, security personnel, or wardens. The alert system complements the mechanical prevention action, providing immediate intervention opportunities. The researchers emphasize the simplicity and cost-effectiveness of their prototype, making it suitable for mass deployment in hostels and rental housing. Unlike camera- or PIR-based solutions, it preserves privacy while maintaining high sensitivity to load changes. Their testing confirmed consistent performance, with negligible false positives when minor movements like dusting or fan cleaning occurred. Limitations include the need for precise calibration and possible sensor drift over time, especially in humid environments. Additionally, the system does not include a local alarm—only remote alerts—which may delay immediate response. Overall, the study represents a practical, privacy-respecting, and affordable engineering solution. The authors propose integrating it into smart home systems with cloud-based monitoring and predictive analytics. The design balances mechanical simplicity with digital communication, forming a foundation for scalable suicide prevention systems adaptable to low-cost environments.

[3].Anti-Suicide Kit Authors: Chidambar P. M. et al. (2024) The Anti-Suicide Kit by Chidambar and co-authors (2024) introduces a robust mechanical fail-safe system that operates independently of electronic sensors or external power. The design incorporates a spring-loaded fan rod engineered to uncoil or extend when subjected to a specific downward force (typically between 20–25 kg). This automatic uncoiling action effectively lowers the fan height, reducing the tension force on the neck and preventing fatal compression. This system's greatest strength lies in its passive reliability—since it operates entirely on mechanical principles, it remains functional during power outages and requires minimal maintenance. In addition to mechanical protection, the kit includes a pressure-based triggering unit connected to an audible alarm siren. When the load threshold is surpassed, the spring expansion simultaneously activates this siren, alerting nearby individuals. Field trials demonstrated remarkable durability and sensitivity, especially in retrofitting existing ceiling fans without major modification. The system's modular nature allows it to be installed within minutes, making it highly suitable for mass adoption in student accommodations or government hostels. However, while it excels in mechanical reliability, it lacks digital connectivity—there is no remote alert feature or data recording mechanism. The authors suggest future integration with smart IoT-enabled nodes to transmit alerts and log incident data for preventive behavioral analytics. By focusing on mechanical innovation, the Anti-Suicide Kit advances a low-cost, maintenance-free safety solution. It contributes 10 significantly to the broader engineering discourse on suicide

prevention by emphasizing sustainable design, passive operation, and user safety, offering a prototype that aligns with both social and technical requirements.

[4].Precautionary Savvy Fan to Prevent Suicide Authors: Kotresh H. M. et al. (2023) Kotresh and colleagues (2023) present a hybrid system that integrates smart sensors with behavioral deterrence mechanisms to prevent suicide attempts via ceiling fans. The proposed “Precautionary Savvy Fan” operates using load-sensing modules linked to an onboard microcontroller that continuously measures weight variations on the fan structure. When a sudden load surpassing the predefined safe limit (typically above 20 kg) is detected, two parallel actions are triggered: (1) the fan’s electronic cut-off and alert system, and (2) the activation of a spray mechanism that releases a non-toxic, harmless irritant or water mist designed to interrupt the suicide attempt immediately. The inclusion of a spray module marks an innovative behavioral intervention strategy that aims not only to disrupt the attempt mechanically but also to disorient the individual long enough for external help to intervene. The fan is also equipped with a wireless notification module that transmits alerts via mobile networks or IoT platforms to wardens, family members, or emergency services. Kotresh et al. emphasize the dual nature of their design—preventive and deterrent—making it effective in both early and active stages of suicide attempts. Their prototype demonstrated reliable triggering and quick response, with latency below two seconds in laboratory trials. While the system introduces notable innovation, potential challenges include refilling of the spray module, long-term reliability of moving parts, and maintenance complexity. Future work aims to miniaturize the design and integrate AI-based load pattern analysis to reduce false activations. Overall, this study expands the paradigm of suicide prevention technology from mere detection to active behavioral disruption, establishing a novel intersection between mechatronics, safety engineering, and psychology.

[5].A Call for Preventing Suicide by Hanging from Ceiling Fans Authors: Kariippanon et al. (2019) Kariippanon et al. (2019) take a multidisciplinary perspective rather than proposing a single engineering device. Their paper acts as a policy framework urging collaboration among engineers, health professionals, psychologists, and policymakers to address suicides by hanging from ceiling fans. The authors identify a critical research gap—most interventions are technical, with minimal integration into systemic public health strategies. The study proposes a three-tier preventive model: (1) Measurement—developing reliable metrics for evaluating fan safety and load thresholds; (2) Design—encouraging manufacturers to incorporate safety mechanisms such as collapsible rods or 11 load-limiting joints; and (3) Evaluation—

implementing standard certification and auditing for suicide resistant fan designs. The authors cite data from institutional environments showing the prevalence of suicides via ceiling fans, particularly in hostels and low-cost housing, arguing that technological innovation must be accompanied by policy enforcement. They recommend that building codes and hostel regulations mandate the use of suicide-resistant ceiling fans, paralleling approaches in public infrastructure safety design. Their approach does not focus on a prototype but serves as a foundation for interdisciplinary research that later inspired practical innovations like those seen in the Kota (2023) implementations. The study also highlights the importance of cultural sensitivity and mental health awareness alongside engineering solutions. In essence, this work reframes suicide prevention from an isolated engineering problem into a systemic societal responsibility, advocating for a comprehensive, data-driven, and ethically guided implementation strategy that combines design innovation, policy, and education.

[6].Real-World Implementations in Kota (India) Authors: Kota Authorities (2023) In 2023, the Kota Municipal Authorities of India implemented a landmark initiative aimed at reducing suicides among students, a demographic particularly vulnerable to academic pressure. The program mandated the installation of spring-loaded ceiling fans in all student hostels and paying guest accommodations. This large-scale intervention represents the first real-world policy translation of prior academic research into municipal-level action. The spring-loaded fans are engineered to automatically expand or collapse when subjected to a load exceeding approximately 20 kilograms, immediately lowering the hanging height and eliminating the potential for fatal neck compression. Additionally, many units include an integrated siren that triggers when the spring mechanism engages, providing an audible alarm to attract attention from nearby individuals. The initiative's success lies in its simplicity and scalability—using purely mechanical designs without reliance on electronics ensures low maintenance and broad applicability in resource-constrained environments. Reports from hostels indicated a sharp decline in suicide incidents following implementation, marking this as a model for preventive engineering at a policy scale. Challenges noted include occasional false activations due to misuse and variability in spring calibration between manufacturers. Nonetheless, the Kota model represents a benchmark for city-level suicide prevention through design policy, bridging the gap between research innovation and real-world deployment. This initiative emphasizes the role of public governance in technology adoption, highlighting how municipal enforcement and engineering design can coalesce into a socially impactful strategy. It establishes a

precedent for other cities to follow, positioning the Kota project as a real-world validation of suicide-preventive mechanical fan designs.

[7].Patent by Dr. Sharma (Jabalpur) Author: Dr. Sharma (2019) Dr. Sharma's (2019) patent marks one of the earliest formalized designs for a suicide prevention ceiling fan. The patented system features a hinged fan rod with a spring-based uncoiling mechanism that disengages when a downward force exceeding 25 kilograms is applied. Upon triggering, the hinge uncoils, effectively extending the fan's length and preventing the exertion of lethal force on the neck. This design's uniqueness lies in its purely mechanical architecture—requiring no power source, sensors, or electronics. The simplicity of the mechanism ensures zero maintenance and longevity, as there are no active components that degrade over time. Moreover, the design allows retrofit installation on existing ceiling fans without requiring a complete replacement, making it both affordable and scalable for mass production. The patent documentation outlines a dual-layer safety structure: the mechanical release prevents fatal hanging, and the uncoiling mechanism automatically reverts to its original position when the load is removed, maintaining usability as a normal ceiling fan. This design later served as the basis for several follow-up studies and implementations, including the Kota (2023) deployment, confirming its engineering soundness. However, it lacks an alert mechanism, meaning intervention depends entirely on mechanical action. Dr. Sharma's innovation exemplifies preventive design philosophy—engineering everyday household objects with built-in safety. His patent is widely regarded as a pioneering contribution that inspired both mechanical and hybrid (electromechanical) anti-suicide systems. Ultimately, this work set the foundation for a new design standard in suicide prevention technology, illustrating how simple mechanical solutions can have profound public health impacts when thoughtfully implemented.

[8].IoT-Enabled Smart Ceiling Fan for Suicide Prevention Authors: R. Karthikeyan & V. Rajasekar (2022) Karthikeyan and Rajasekar (2022) introduce an IoT-driven suicide prevention system integrated into a ceiling fan that combines load detection, motion sensing, and real-time data communication. Their design centers around an ESP32 microcontroller, chosen for its Wi-Fi and Bluetooth capabilities, connected to a load cell, ultrasonic proximity sensor, and servo motor. The fan continuously monitors the load exerted on its blades or rod. When a sustained weight exceeding 20 kg is detected, the microcontroller triggers two actions: (1) an automatic rod-lowering mechanism driven by the servo motor, and (2) real-time notifications via the Blynk IoT platform, sending alerts to connected smartphones and cloud dashboards. This approach allows instant intervention while maintaining data logs for

behavioral monitoring. The use of cloud storage enables long-term data analysis to detect recurring patterns, potentially predicting suicide risk behaviors before they escalate. 13 The authors also included voice alerts through a connected speaker, designed to dissuade the individual in the moment. Testing showed consistent triggering accuracy and stable connectivity, with a delay of less than 2.5 seconds from detection to mechanical response. However, challenges include network dependency, energy consumption, and privacy concerns related to continuous monitoring. This study bridges engineering, IoT, and mental health technology, proposing a scalable model that could integrate with smart home ecosystems or institutional monitoring platforms. It marks a transition from purely mechanical systems to connected, data-driven prevention architectures, enabling both reactive and predictive safety mechanisms.

[9]. Smart Monitoring System for Suicide Prevention Using IoT Authors: A. Kumar, R. Singh (2021) This study proposes an IoT-based monitoring system designed to detect abnormal activities related to suicide attempts and send real-time alerts to caregivers. The system integrates sensors with a microcontroller and wireless communication modules to continuously monitor environmental conditions and unusual events. When abnormal behavior or sudden load changes are detected, the system immediately triggers an alert notification through a mobile application or cloud platform. The proposed system emphasizes the importance of real-time monitoring and rapid communication to reduce the delay in providing help. By using IoT technology, the system allows remote monitoring of individuals in sensitive environments such as hostels, hospitals, and elderly care homes. The research highlights that integrating sensing devices with communication technologies can significantly improve emergency response time. However, the authors also mention that the system requires stable internet connectivity and proper calibration of sensors to avoid false alerts. Future improvements suggested include integrating additional sensors and improving data analytics for better prediction of risky situations.

[10]. Intelligent Safety Ceiling Fan for Suicide Prevention Authors: P. Ramesh, S. Narayanan (2022) This research presents the design of an intelligent safety ceiling fan that incorporates mechanical safety features along with electronic monitoring. The system uses load sensing techniques to detect abnormal weight applied to the fan structure. When the load exceeds a predefined threshold, the system activates an alert mechanism and sends notifications to responsible authorities. The design also focuses on modifying the fan mounting structure to prevent the fan from supporting excessive weight. In addition to the mechanical modification,

electronic components such as microcontrollers and communication modules are used to generate alarms and remote alerts. The authors highlight that combining mechanical safety with electronic communication improves the effectiveness of suicide prevention systems. The proposed design is considered suitable for installation in hostels, hospitals, and residential buildings where early detection and quick response are critical.